

Using Hierarchical Models to Estimate Heterogeneous Effects

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Abstract

Conditional theories are ubiquitous in political science, but standard interaction models cannot handle this proliferation of theoretically informed modifiers. This paper describes why, when, and how to use hierarchical models to encompass multiple sources of heterogeneity and regularize estimates. Doing so is simple, and starts with identifying the key sources of heterogeneity and grouping observations accordingly. Next, fit a hierarchical model with two linked regressions, one connecting treatment with the outcome, and another that models the treatment effects with potential sources of heterogeneity and partially pools group estimates. This captures systematic and random variation in heterogeneous effects, encompasses many interactions, and fits commonly used modeling frameworks, but regularization in this model breaks down if there are many small groups. I document these claims with a simulation analysis and extension of a published study.

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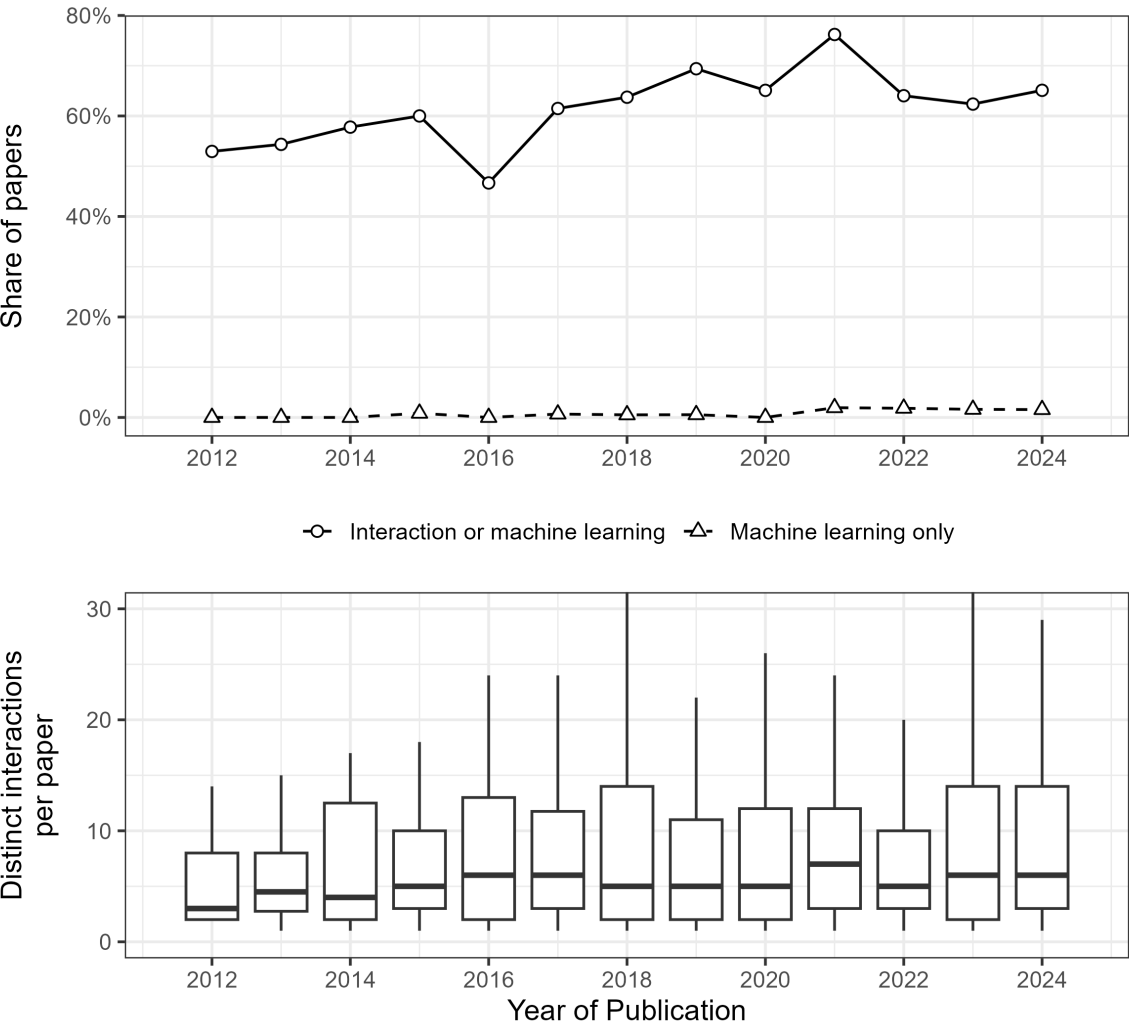
Social scientists often propose conditional theories (Clark and Golder, 2023) and are interested in how different units respond to the same stimulus for theoretical, normative or policy reasons. While most theories consider a single modifier, many theories propose distinct modifiers for the same independent variable. Therefore, multiple modifiers are common. For example, international relations experiments on audience costs (Tomz, 2007) have considered how foreign policy dispositions (Kertzer and Brutger, 2016), partisanship (Levendusky and Horowitz, 2012), gender (Barnhart et al., 2020; Schwartz and Blair, 2020) and policy preferences (Chaudoin, 2014) modify reactions to a leader backing down from a threat. Other audience cost experiments manipulate casualties (Davies and Johns, 2013), how the leader backs down (Lin-Greenberg, 2019), other elite reactions (Levendusky and Horowitz, 2012) and promises vs threats (Levy et al., 2015).

Many potential modifiers are evident in how scholars conduct quantitative research. In the *American Political Science Review*, *American Journal of Political Science* and *Journal of Politics*, many researchers estimate some modifier of treatment effects, either to test their argument or explore others. Figure 1 documents the prevalence of heterogeneous effects estimates from interactions or machine learning in Harvard Dataverse replication data from the *American Political Science Review*, *American Journal of Political Science*, and *Journal of Politics* from 2010 to 2024. In this figure I count the number of papers that fit at least one interaction or machine-learning model of effect heterogeneity in their replication data to show the prevalence of these tools. I did this by downloading all quantitative replication materials hosted in the Harvard dataverse, and then measuring the prevalence of heterogeneous effects estimates. I then used an open-weights LLM (gpt-oss-120b), to parse the code and identify models and interactions.

Interactions are very common. In most years, 60% of papers in the APSR, AJPS, and JOP estimate at least one interaction. Use of machine-learning tools for heterogeneous effects is much less common, and is usually in the single-digits.¹

¹As another indicator of interaction popularity, Brambor, Clark and Golder (2006) has 8,279 citations as of

Heterogeneous Effect Estimation in Leading Journals
2012-2024



Lower panel covers papers estimating at least one interaction and drops outliers from view

Figure 1. *Prevalence of heterogeneous effect estimates, 2012 to 2024.*

Moreover, researchers fit many interactions, as the median paper estimates 6 distinct interaction models. A few outliers estimate as many as 50. Most applied research relies on parametric interactions, with different theories and robustness checks leading to multiple interactions in most papers.

Unfortunately, the workhorse regression interactions that many scholars rely on are ill-suited to test the growing number of conditional arguments in social science. Scholars cannot ignore heterogeneity, but the most common tools either increase the risk of spurious results or are hard to interpret and use. Interaction terms and subgroup analysis are ubiquitous because they are relatively easy to interpret, but they increase the risk of spurious results. Many political science analyses have low power even to detect effects without interactions (Arel-Bundock et al., 2026). Adequate power to estimate even a single interaction can require significantly more data (Gelman, 2018), which may be prohibitively expensive or impossible. As a result, statistically significant heterogeneous effect estimates may be far too large—the result of noise in the data, not systematic differences. This problem is partially responsible for widespread issues replicating findings based on interactions (Simmons, Nelson and Simonsohn, 2011). Adding more modifiers makes interpretation harder and increases these power problems. Researchers might instead fit separate pairwise interactions for each modifier, but multiples models and corresponding testing problems.

Given multiple sources of heterogeneity, machine-learning tools such as random forests (Green and Kern, 2012; Wager and Athey, 2018), support vector machines (Imai and Ratkovic, 2013), and ensemble methods (Grimmer, Messing and Westwood, 2017; Künzel et al., 2019; Dorie et al., 2022) are more likely to avoid over-fitting. These machine learning algorithms usually have some regularization component and can discover complex patterns and high-dimensional variation across multiple modifiers.² These tools can be difficult to interpret and

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²LASSO models sometimes address similar issues (Ratkovic and Tingley, 2017). Blackwell and Olson (2022) describe a lasso approach to interactions of a single modifier and all the variables in a model that sits between

implement, however, especially in smaller social science datasets. The relative lack of theoretical guidance and interpretability is especially problematic for testing the relative weight of multiple conditional arguments. If researchers want to compare the impact of different modifiers, machine learning models use quantities that many applied researchers are less familiar with, so these techniques are far less popular than interactions.

Despite a large statistics literature documenting how regression-based interpretation and regularization make Bayesian hierarchical models a useful tool for heterogeneous effects estimation (e.g., Feller and Gelman (2015); McElreath (2016); Dorie et al. (2022)), political scientists rarely use this tool. For instance, of the three applied political science citations of Feller and Gelman (2015), only Marquardt (2022) models treatment effects. The MAIDHA (Multi-level Analysis of Individual and Heterogeneity and Discriminatory Accuracy) framework in intersectionality research is a noteworthy exception (e.g. Much, 2025). These researchers estimate distinct effects for groups defined by class, race and gender, and partial pooling helps give reasonable estimates for small subgroups in survey data.

This paper provides practical guidance by explaining how, when and why to use hierarchical models of heterogeneous effects. The hierarchical varying slopes strategy I advocate here addresses the power shortcomings of interactions while allowing more theoretical guidance and comparison than machine learning. It does so by regularizing estimates of multiple modifiers, thus trading some bias of specific effects for reduced variance in effect estimates. Taking this side of the bias-variance tradeoff improves accuracy overall, and reduces errors about the sign and magnitude of non-zero effects.

Using hierarchical models is more flexible than standard interactions but easier to implement and interpret than machine learning. It preserves a straightforward structure while accommodating more factors and ameliorating the downsides of subgroup analysis. The main

machine learning and linear regressions. This considers a single interaction of interest and bias from other unmodeled interactions with controls.

weaknesses are that unlike machine learning, the hierarchical approach lacks the flexibility to discover high-dimensional heterogeneity and regularization will break down with many small groups. It can also be slow to fit on large datasets, depending on the underlying sampler.³

There are two key steps when theory and data make using hierarchical models worthwhile. First, researchers should identify potential modifiers of a treatment effect and use them to group observations and model treatment effects. Second, they should take that model of treatment effects and connect it to a model linking the treatment and outcome, implementing a type of varying slopes model. Modeling heterogeneous effects produces interpretable results, which facilitates argument testing. It also allows researchers to compare different sources of heterogeneous effects and describe effect variation. These are crucial advantages in a world with many conditional theories.

While frequentist estimation of hierarchical models is possible, Bayesian estimation is easy, more informative and more robust (Stegmueller, 2013). Bayesian estimation provides crucial information by connecting parameters through common prior distributions, thereby regularizing estimates and propagating uncertainty. Working with posterior distributions also gives researchers more flexibility to describe how and when effects vary and use prior predictive checks to examine the assumptions in their models. While computation and coding were once a barrier to employing Bayesian methods, fitting a wide range of hierarchical models is straightforward with the `brms` package in R (Bürkner, 2017).⁴

In the remainder of this paper, I start by describing how and when to estimate hierarchical models of heterogeneous effects. I then use simulations to compare OLS and hierarchical estimates of treatment effects under different conditions. Finally, I demonstrate the process by analyzing a study of how military alliances shape public support for war by Tomz and Weeks

³Goplerud (2021) considers a model that uses Bayesian structured sparsity to estimate which group coefficients are similar and which are different. In this approach, researchers use theory to inform potential groups, but the data determines common estimates for groups.

⁴I provide example code below and in the appendix.

(2021). The reanalysis documents the importance of regularization in analyzing subgroups derived from combinations of experimental treatments.

1 Hierarchical Modeling of Heterogeneous Effects

There are two steps in hierarchical models of heterogeneous effects. This is simply a different way of expressing varying slopes models. First, researchers must identify potential sources of heterogeneity, and think about the right model of heterogeneity. This will also depend on what variation is most important and interesting. Theory, policy concerns, or normative factors are all possible motivations.

This first step determines what heterogeneous effects a researcher estimates. It is analogous to researchers thinking through a regression specification and requires similar care. Researchers need to define what variation is most important, link heterogeneous effects to theory, and structure modeling. This is especially important for pre-registration, and could reduce the amount of exploratory analyses in registrations. Not thinking carefully about sources of heterogeneity will obfuscate results and can hinder model fitting.

There are three general approaches to defining key modifiers. First, researchers can use combinations of other treatments, especially when an intervention has several dimensions but theory emphasizes one of them. The experimental design determines modifiers, and the model estimates heterogeneous treatment effects. If researchers want to know how different issues shape the impact of elite foreign policy cues (Guisinger and Saunders, 2017), they could include indicators of issues, for instance. A similar application of hierarchical estimators for topic-sampling experiments estimates how a treatment effect varies across topics (Clifford and Rainey, 2024).

At the moment, researchers often estimate the impact of different treatment combinations with fully crossed interactions. This estimates the impact of a treatment across experimental

strata, but risks spurious results by functionally estimating subgroup results. These estimates are often noisy however, and can rely on 50 or fewer data points in smaller samples.

A second approach uses unit, demographic and contextual factors to estimate effect heterogeneity. Here, researchers examine what factors within or around units shape their response to an independent variable. Researchers could use a mix of individual and contextual factors to predict divergent consequences of a survey experiment treatment. Such a model might include factors such as an indicator of state of residence, age, gender, and race.⁵ For example, Alley (2021) uses alliance characteristics to examine when alliance membership increases or decreases military spending.

Third, researchers might use hierarchical models to address specific policy concerns. Policy analysts often want to know how an intervention impacts a specific population. Researchers might want to know if a job-training program improves employment outcomes for black women in the South, for instance. To do this, a researcher might specify a heterogeneous effects model with race, gender and region, plus additional controls or other factors.

After defining moderators and how they relate to treatment effects, the second step is fitting a hierarchical model that links a model of the outcome with a model of heterogeneity. Essentially, researchers model the outcome and the process that produces heterogeneous treatments. The model employs two connected regressions. One regression deals with the outcome, and includes the treatment effects.⁶

The other regression models the treatment effects. In this regression, theory should inform which modifiers predict the treatment effect. To regularize the estimates, this second equation must include varying intercepts for groups within the data. Groups can be based on combinations of discrete modifiers and binned continuous modifiers. As a result, this model includes

⁵Extrapolations to a representative sample of a subpopulation might require poststratification.

⁶If other units such as states define the groups, rather than combinations of modifying variables, then adding group-level predictors to this equation is essential. For example, in a model where an effect varies by state, adding state-level variables like ideology, population and GDP would avoid partially pooling small groups too far towards the overall mean.

both systematic and random variation in the impact of a key independent variable on different groups.

To maximize variation and preserve regularization, researchers should use continuous modifiers directly in the regression and bin them for setting groups. Not binning modifiers will create many small groups, leading the hierarchical component to break down. In the appendix, I document some additional best practices for continuous modifiers, including avoiding smooth priors for ordered effects across bins and the importance of bin boundaries relative to the number of bins.

Setting groups is a critical task that requires balancing detail against regularization. Larger groups can deviate more from the overall mean, as their data contributes more to the posterior than the prior. A few small groups means less need for a hierarchical model. Smaller groups will be shrunk more towards the overall mean. Too many small groups and the regularization component will have less information to work with. Using model fit criteria such as WAIC or leave-one-out cross validation can help guide group formulation. Pre-registration is also crucial, because this the area with the greatest researcher degrees of freedom.

Should the variables in the heterogeneity equation also appear in the outcome equation? Here, the guidance of Brambor, Clark and Golder (2006) to include constituent terms in an interaction will often apply. If a modifier is correlated with treatment and the outcome, then it is also a useful control, as it has an independent relationship with the dependent variable in addition to modifying the treatment effects. The functional form of the two equations may differ, however.

I now briefly describe a generic hierarchical model.⁷ For ease of exposition, consider making between-unit comparisons based on an experimental treatment. Start with N units indexed by i , some of which receive a binary treatment T . Assume that the outcome variable y is nor-

⁷I split the equations for ease of exposition, but it is straightforward to see that this approach places priors on an additive interaction model of the form $y = T * (z_1 + z_2 + z_3...) + \mathbf{X}\beta$

mally distributed with mean μ_i and standard deviation σ .⁸ g indexes researcher-defined groups based on combinations of the modifying variables.

The outcome for each unit depends on an overall intercept α , an optional matrix of control variables $\mathbf{X}$, and a set of group treatment effects λ_g . When T is binary, estimated λ parameters for untreated units have no impact on the outcome and drop out of the likelihood. For a continuous treatment, effects will depend on the product of T_i and λ .

$$y_i \sim N(\mu_i, \sigma) \quad (\text{Likelihood})$$

$$\mu_i = \alpha_g + \lambda_g T + \mathbf{X}\beta \quad (\text{Outcome Equation})$$

$$\lambda_g = \theta_g + \mathbf{Z}\gamma \quad (\text{Heterogeneity Equation})$$

$$\theta_g \sim N(\mu_\theta, \sigma_\theta) \quad (\text{Group Varying Intercepts})$$

The heterogeneity equation then models those group treatment effects with a systematic and random component. The systematic component is a matrix of predictors $\mathbf{Z}$ and associated parameters γ . Z can mirror any regression specification researchers might use for an outcome; linear combinations of variables, interactions, smooth functions. Interactions of the modifiers could capture processes where combinations of modifiers produce non-additive jumps in heterogeneity, for instance.

The random component of the heterogeneity equation is a series of group-specific varying intercepts θ_g . These are critical, because they capture group deviations from the systematic trends expressed in the design matrix $\mathbf{Z}$. Group outliers in the λ estimates are partially pooled back towards the overall mean μ_θ . The variance parameter σ_θ controls the dispersion of the group effects around the overall mean.⁹

⁸Researchers can and should use binary, categorical and other outcome likelihoods.

⁹In brms, using non-linear syntax can express a model with a treatment, two controls, and three modifiers as: $y \sim \text{lambda} * \text{treat} + \text{controls}, \text{lambda} \sim \text{mod1} + \text{mod2} + \text{mod3} + (1 | \text{mod1} + \text{mod2} + \text{mod3}), \text{controls} \sim \text{control1}$

The above model can be fit with Bayesian or frequentist methods, but Bayesian estimation offers important advantages. First, it is more flexible, and including prior information can facilitate model fitting. Priors on the α , β , and γ parameters will matter more in very small datasets. The more important prior is for σ_θ , and depending on beliefs about potential dispersion across groups, half-normal or half student-t priors are common choices. This prior is what helps regularize estimates by pulling extreme groups towards the overall mean μ_θ .

Bayesian modeling offers additional information and a wealth of model-checking tools. STAN and brms, the software I provide guidance for here, facilitates prior predictive checks, where researchers simulate from the model to check what estimates are plausible with specific priors. Working with posterior distributions then provides a wealth of information about effect heterogeneity and propagates uncertainty. Last, Bayesian models can provide useful diagnostics such as divergent transitions that can be warnings of specification problems. It is often better to modify a model with fitting problems instead of brute-forcing it through the sampler.

Depending on the problem, researchers can extend this initial model in many ways. The generative core of Bayesian hierarchical models adds significant flexibility. For instance, they might model the variance of the treatment effects or the outcome. Fitting splines or smooth functions is also possible, or researchers might extend the model in continuous group space with Gaussian process regression (Prati et al., 2023).

In interpreting these estimates, researchers should leverage the full range of information from the different parameters. First, the λ posteriors give the impact of the treatment on each group, and are the core quantity of interest. All λ s reflect a systematic component from the predictors in $\mathbf{Z}\gamma$ and random variation from θ . γ parameters can, depending on the regression, be interpreted as the impact of a change in a modifier on the treatment effects. For example, a γ of .1 on a binary modifier means that λ is .1 higher in expectation when the modifier is

+ control2, nl = TRUE

one, and .1 lower when it is zero. σ_θ thus measures treatment variation outside the systematic regression. Other techniques such as interactions in OLS with robust standard errors provide less information.

2 Performance on Simulated Data

To assess how this hierarchical model compares to interactions, I first assess their performance on simulated data. I do not include machine-learning tools in the simulation because they often produce different effects, such as individual-level estimates that are not directly comparable to the group-level hierarchical and interaction estimates. I undertake a few illustrative comparisons in the appendix, however.

In the simulations, I fix the number of observations, and then generate different scenarios with distinct groups and distributions of true effects across those groups. All of these scenarios give a group-specific effect of a binary treatment variable. I then estimate those effects with three models. First, ordinary least squares where the treatment interacts with the group variables additively. Second, ordinary least squares with fully crossed interactions between the grouping variables. The fully crossed model is somewhat extreme, but it appears regularly in applied research, especially for subgroup effects in factorial experiments.¹⁰ Third, a hierarchical model where an additive combination of grouping variables predicts treatment effects in each group and partial pooling regularizes effect variance. In the hierarchical model, I use main effects of the group variables in the heterogeneity equation and set a random intercept for each group. Fully crossing the group variables would saturate the model and make the random intercepts redundant.

There are seven total data-generating processes. One uses equal group sizes and normal effect distribution, which matches the partial pooling process well. Another uses heavy tails,

¹⁰The application below takes this approach.

drawing effects from a t-distribution with three degrees of freedom. A second creates imbalanced groups with the normal effect distribution, because larger groups have more information and can thus diverge more from the overall mean, while partial pooling shrinks smaller groups closer to the overall mean. To further test this mechanism, I implement another process where larger groups have larger effects, so partial pooling takes small groups closer to a potentially unrepresentative mean. Another tests include a sparse effect distribution with only a few different estimates, a bimodal distribution, and one with systematic effect variation that might be especially friendly to the additive interactions.

When I divide the 2,000 observations into groups based on combinations of three to seven binary variables and give equal probability of assigning observations to a group, there are between 8 and 128 unique groups. This gives approximately 250 observations per group at the low end and 16 observations at the high end. The imbalanced simulation can have hundreds of data points in one group and as few as two in another, but I fix the data-generating process so that all groups populate.

I then compare the models on root-mean squared error (RMSE) and the rate of type M and type S errors. The root-mean squared error of the coefficient estimates compared to the true value is the most general metric, and is the best because the hierarchical model's shrinkage will introduce bias but compensate with lower variance (Clifford and Rainey, 2024). The bias-variance trade-off is the core of the hierarchical advantage. Lower variance may overcome greater bias in specific coefficients and make the hierarchical model more accurate on average. Pooling to the treatment mean is conservative in some ways, but that conservative approach is useful when we never know the true distribution of treatment effects. RMSE also captures instances where an estimator exaggerates the magnitude of the coefficient.

I summarize the RMSE of the hierarchical and OLS models in Figure 2. All three models perform worse as the number of groups increases and each group becomes smaller and noisier. OLS with fully crossed interactions of grouping variables performs comparably to the hier-

archical model when there are 8 groups, but RMSE increases dramatically as the number of groups increases. Unequal group sizes make this problem especially acute.

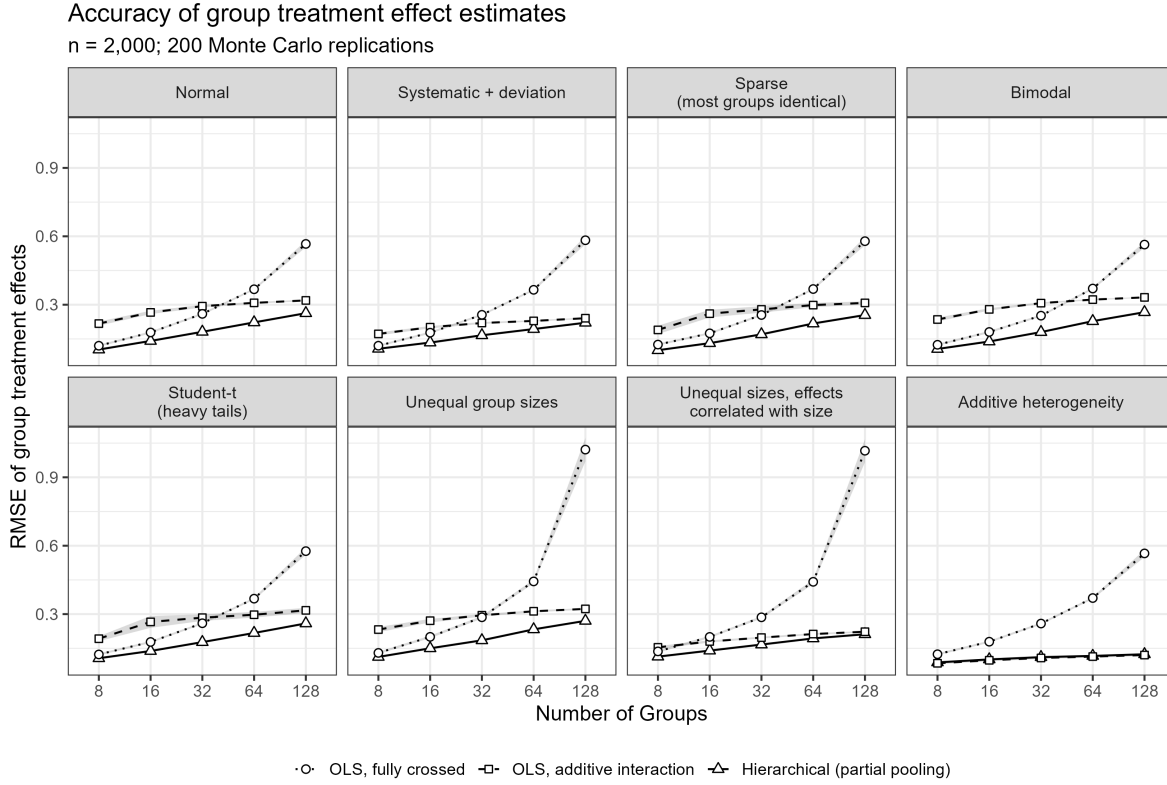


Figure 2. Comparison of OLS and hierarchical root mean squared error in group coefficient estimates. Simulation fixes the sample size at 2,000 observations and varies the number of groups. Each group has equal assignment probability.

The additive OLS specification does not decay in the same way. For some group configurations, there is little difference as group size increases. The downside is consistently larger RMSE relative to the hierarchical model across all of the group sizes and configurations.

The hierarchical model is usually best in RMSE, especially as groups proliferate. With few groups, some OLS models perform marginally better, but not clearly so. Like most models, accuracy degrades as groups proliferate, because there is greater scope for bias.

These patterns are the result of reduced variance overcoming greater bias in the hierarchical estimates. I illustrate this pattern in Figure 3, which draws from the simulations 8 and

128 groups with a normal effect distribution for ease of presentation. This figure plots the true against estimated values in those simulations for the three model specifications, using the normal distribution of true effects to illustrate. To clarify the pattern, I regress the true values on the estimated values.

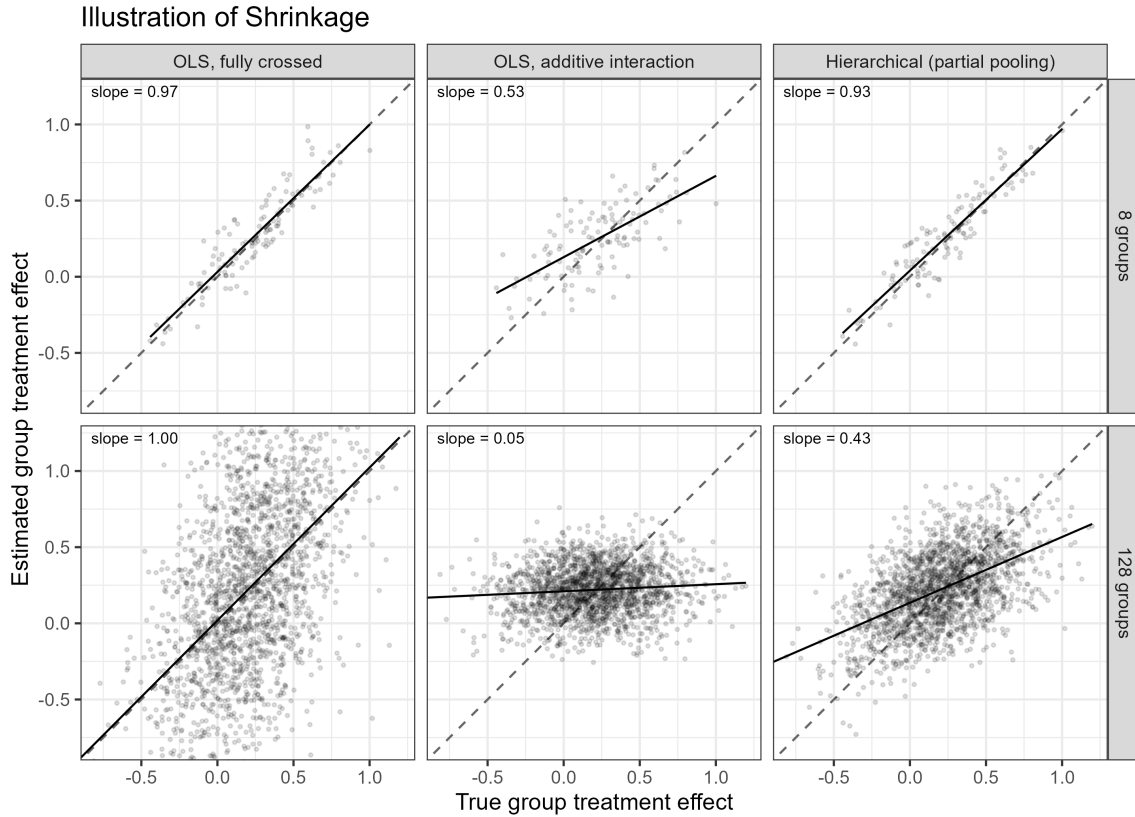


Figure 3. True and estimated effects from simulations with 8 and 128 groups. The black line regresses the true value on the estimates. Points on the dashed line are unbiased

Under 8 groups, the hierarchical models and fully crossed OLS have similar slopes, with marginally less variance around the true values for the hierarchical model. This is why these two specifications have similar RMSE with a few groups. OLS with an additive interaction of treatment and modifiers gives a single common answer, smoothing out too much of the true variation.

At the other extreme of 128 groups, fully crossed interactions have significant noise around

the true value. This variance increases RMSE, despite less bias in the overall trend. The additive model flattens out the overall trend, giving a single trend. It also has noise, but far less than the fully-crossed OLS, so the RMSE is much better. The hierarchical model falls in between. It has more bias than fully crossed OLS, but much lower variance. It recovers the true values better than the additive OLS specification, with somewhat more variation.

This is the bias-variance tradeoff in action. Hierarchical models are not as unbiased as some specification, but the reduction in variance makes them more accurate on average. This conservative approach is especially important when there are many modifiers and group size decreases.

Conservative estimation of subgroup effects is especially important when researchers use statistical significance thresholds. Hierarchical modeling reduces the risk of that statistically significant results have the wrong sign or an exaggerated size (Gelman and Carlin, 2014). These Type S (sign) and Type M (magnitude) errors are just as important as whether an uncertainty interval includes zero. Misstating the direction of an effect misleads in theory testing, and magnitude errors fail to capture the substantive importance of an effect.

Figure 4 shows the rates of Type S and M errors in estimates where the 95% confidence or credible interval excludes zero.¹¹ This figure plots error rates at 8, 32, and 128 groups. The rate gives the percentage of estimates with a sign error and the exaggeration ratio for Type M errors.

With sign errors, hierarchical models and fully crossed OLS perform similarly. The additive OLS model makes more sign errors because it fails to pick up adequate variation, especially as the number of groups increases. In some cases, as many as 1/3 of statistically significant effects are the wrong sign using additive OLS interactions. Other models make errors too, but they are far more rare.

Fully crossed OLS has more problems of effect magnitude, as statistically significant effects

¹¹Some OLS estimates drop due to perfect collinearity and are excluded.

Errors of inference among subgroup effects declared significant
Intervals excluding zero: posterior for the hierarchical model, delta method for OLS.

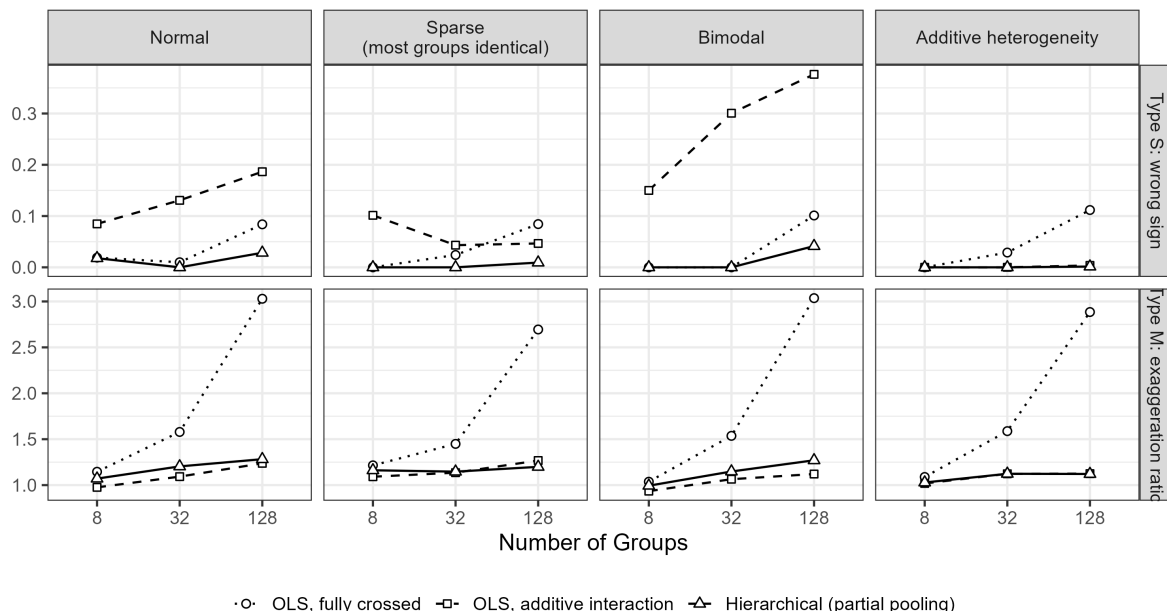


Figure 4. Type M and S error rates.

are often large overestimates. With small groups and substantial noise, the exaggeration ratio approaches 3. Additive OLS and hierarchical models are less likely to exaggerate.

The simulation therefore shows the virtues of hierarchical modeling in three steps. In the aggregate, hierarchical models outperform the most common OLS specifications because their estimates are less variable, although slightly more biased due to partial pooling. This reduces the rate of misleading conclusions about the sign and magnitude of non-zero effects.

3 Example Application: Alliances and Public Support for War

In the following, I demonstrate how the hierarchical approach works in practice and compares to prior applied work by reanalyzing a study by Tomz and Weeks (2021). Tomz and Weeks (TW hereafter) examine whether the public is more willing to go to war for when the

beneficiary of that intervention is a U.S. ally. In a factorial experiment with vignettes, they find a 33% average increase in support for military intervention on behalf of allies, compared to non-allies. This is a large and potentially important relationship because the United States has a global network of allies.

Given the size of the main effect, TW's paper is an ideal scenario for comparing interactions and hierarchical models. Corresponding interaction effects may be large, and their sample size of 1,200 respondents is not unusual in published work. At the same time, TW estimated many interactions to check how other treatments modify the impact of alliances by fully crossing interactions of their experimental treatments.¹² There are 64 unique treatment groups with anywhere from 11 to 32 respondents, so estimates of the impact of alliances in the 32 pairs of alliance treatment and control groups employ at most 54 data points. As such, hierarchical regularization will likely change some conclusions by pooling noisy estimates in small groups. For ease of comparison, I also compare an OLS model with additive instead of fully crossed interactions. As the simulation results indicate, the fully crossed results are far noisier.

3.1 Differences by Experimental Scenario

Along with alliances, TW randomly assign whether the potential beneficiary of U.S. intervention is a democracy or not, the stakes of intervention, the potential costs, and the region of the world. They estimate the impact of alliances in the 32 treatment conditions with an OLS model that fully crosses interactions between the treatments, and calculate marginal effects over different averages of these groups. I use a hierarchical model to estimate the impact of alliances, with the experimental treatments as systematic modifiers of the alliance effect and group-specific intercepts. I keep their Gaussian likelihood to minimize changes, although a logistic regression would fit the data better.¹³ The resulting approach mirrors TW's model,

¹²This is one example of fully crossed interactions in applied research.

¹³In the appendix, I show that a Bernoulli likelihood does not change the results.

but regularizes the estimates with partial pooling.

Comparison of Heterogeneous Treatment Estimates

Heterogeneity from Experimental Conditions

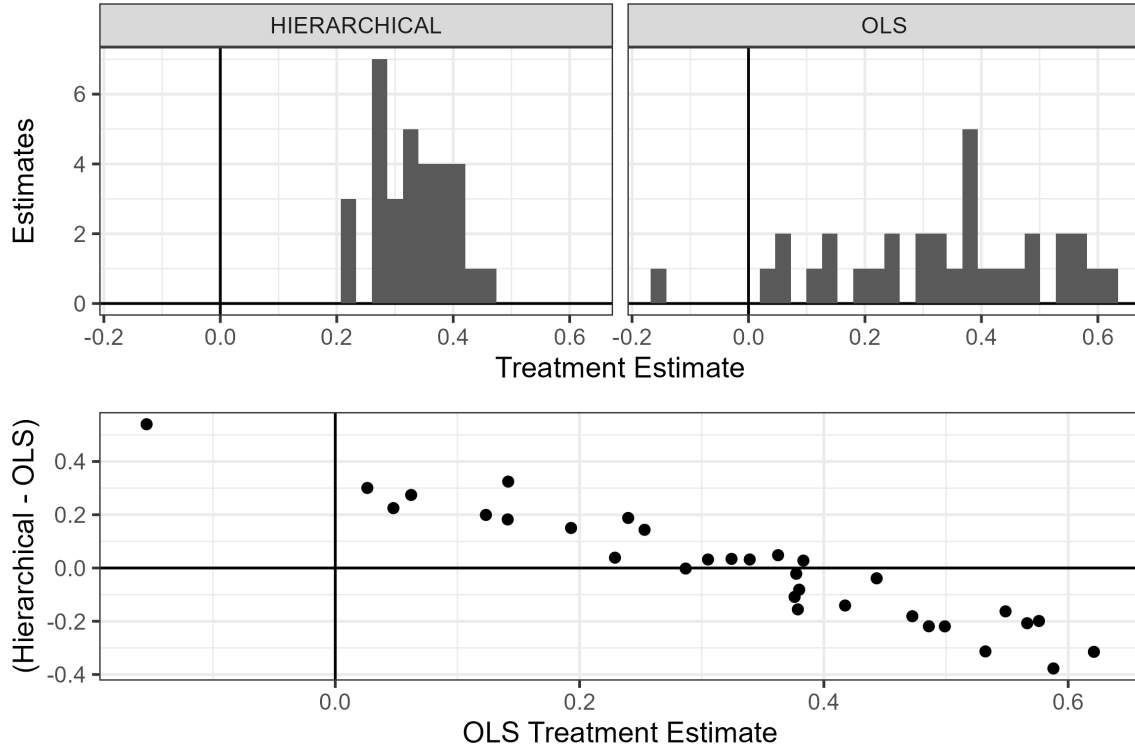


Figure 5. Comparison of OLS and hierarchical estimates of the impact of alliance across experimental conditions. The top panel gives a histogram of the treatment effects from each model. The bottom panel gives the difference between the hierarchical and OLS estimates for each group.

Figure 5 compares the estimated alliance treatment effects across the experimental groups with the OLS and hierarchical models. This figure illustrates the regularization of hierarchical modeling in two ways. It shows the difference between the median hierarchical estimate and the OLS estimate for each combination of treatment groups and plots the distribution of effects.

First, the hierarchical estimates are much less variable and more tightly clustered around the overall mean. This occurs because the OLS model mechanically forces treatment effect estimates to the value of the coefficients and corresponding subgroup means. Given the size of this sample, the treatment effects are based on comparisons of roughly 25 treatment and 25

control respondents.

Second, the hierarchical estimates pull in unusual values. The bottom panel of Figure 5 shows the difference between the hierarchical and OLS estimate for each group. The downward slope from positive to negative values means that the hierarchical estimates are larger than below average OLS estimates, but smaller than above average interaction estimates. The hierarchical estimates shift towards the overall mean, away from extreme values. The OLS model finds one scenario where alliances reduce support by almost 20%, but the hierarchical model treats this as noise and brings it towards the grand mean, leading to a 50% difference.

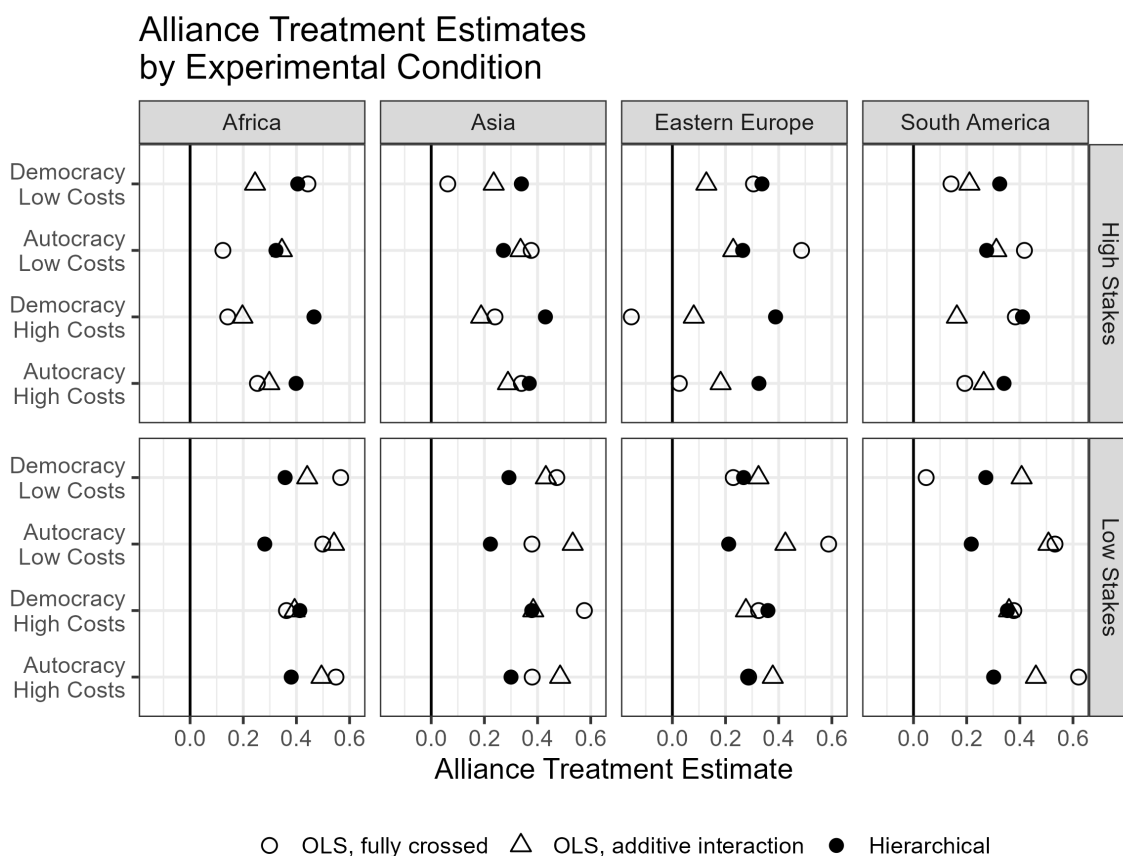


Figure 6. Comparison of OLS and hierarchical estimates of the impact of alliance across experimental conditions. Each point gives the effect estimate for each group.

I compare the estimates for each group with greater detail about what shapes each hetero-

geneous treatment in Figure 6. This clearly shows the reduced variation in the hierarchical estimates across the different experimental conditions. In some cases, the hierarchical and OLS estimates are very similar, but the model suggests that in a smaller sample, saturated interactions of treatment effects overstate effect heterogeneity. For some conditions, such as a low-stakes, low-cost intervention to support a South American autocracy, the hierarchical estimate is 30% lower than the OLS estimate. The condition where TW find a 20% drop in support for an ally compared to a non-ally is a high-cost intervention on behalf of an Eastern European democracy. That would be concerning, but the hierarchical model suggests that this is closer to a 30% increase. The 30% increase matches later findings on how randomly assigning NATO membership to different Eastern European states impacts U.S. public support for war (Tomz, Weeks and Bansak, 2023).

Given the very large average effect of alliances, these results do not change the direction of the findings- in some scenarios they even strengthen them. However, the magnitudes of some subgroup estimates do matter, as they might lead observers to misidentify when alliances matter most. The hierarchical models suggest that low stakes are not as influential as the OLS results indicate, for example. For other, smaller effects, regularization with a hierarchical model might change inferences in subgroup analysis.

In the appendix, I document three patterns. First, I check sensitivity to the likelihood. I also check whether covariate adjustment explains the results, instead of partial pooling. Last, I consider the standard deviation on the variance hyperparameter and show that only one group has opposite signs.

These results show some potential uses of the hierarchical approach to heterogeneous effects.¹⁴ Regularization moderates what might otherwise be extreme inferences about experimental subgroups, and as I show in the appendix, provides useful inferences about demographic heterogeneity.

¹⁴In the appendix, I analyze Bush and Prather (2020) and detail additional heterogeneous effects from TW.

4 Conclusion

As conditional theories proliferate in social science, many independent variables have multiple potential modifiers. Interaction models cannot easily handle this shift, as adding three or more variables to an interaction reduces the interpretation benefits and fitting separate models for each modifier increases the risk of spurious results. Hierarchical modeling retains the clear interpretation of interactions in a single model that regularizes estimates.

I have therefore explained how to use hierarchical models to estimate heterogeneous effects. The first step first identify sources of heterogeneity, usually unit characteristics or other treatments, and set corresponding groups. Then, fit two regressions, one that models the relationship between a treatment and the outcome, and another that models the treatment effect with the modifiers and varying intercepts by group.

Explicitly modeling how different groups respond to an independent variable can help test arguments and inform policy. It also details what drives variation in an effect and how much an effect varies. By using partial pooling of different groups, hierarchical modeling is less vulnerable to spurious inferences about subgroups.

The main downside of this model is that as a researcher adds more groups and group size shrinks, partial pooling becomes less useful. Small groups lack sufficient data to offset the prior, and are thus pooled strongly to the overall mean. If all the groups are small, this reduces the estimated variance in effect sizes so much that in extreme cases, it will collapse to basically zero. At this point, the systematic component in the regression will dominate and produce conclusions that are closer to interaction models.

Hierarchical modeling provides an intermediate approach between interactions and machine learning algorithms. For interactions with one or perhaps two modifiers, relying on simple interaction tools is best. Machine learning is best for discovery of complex heterogeneity, especially in large data. When there are two or more theoretically informed modifiers

and researchers want to test their relative weight, hierarchical modeling allows flexible and interpretable estimation of effect variation.

Future work can build on this approach in other ways. Gaussian process regression is another flexible estimation tool with similar benefits that could extend to similar problems (Prati et al., 2023). Considering temporal, spatial and spatiotemporal autocorrelation in both equations is another worthwhile task. Another interesting application might be pooling experimental data and using the hierarchical structure to compare different manipulations in experiments with similar structure and smaller modifications. These tasks would further expand the benefits of hierarchical modeling in tackling multiple conditional arguments in a single framework.¹⁵

¹⁵AI Disclosure: I used Claude Code to refine simulation code and conduct the analysis of prior publications encapsulated in figure 1. I also checked the all the code for bugs or problems with Claude Code.

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